

Digital Image Processing

Lecture # 1 **Introduction & Fundamentals**

Contents

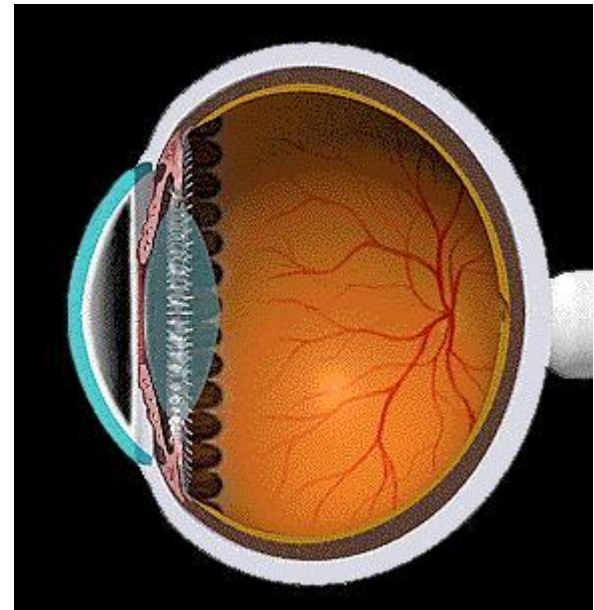
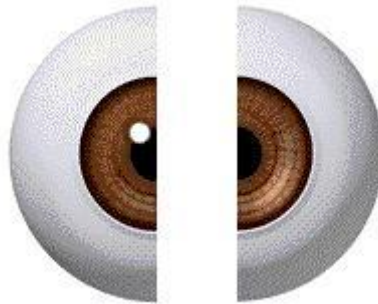
- Human Eye
- Image Perception
- Digital Images
- Image formation

Human and Image Perception

- We can't think of image processing without human vision system.
- We observe and evaluate the images that we process with our visual system.
- Without taking this elementary fact into consideration, we may be much misled in the interpretation of images.

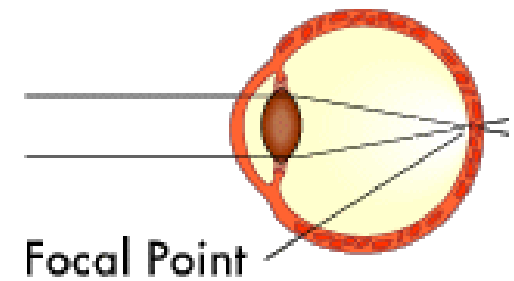
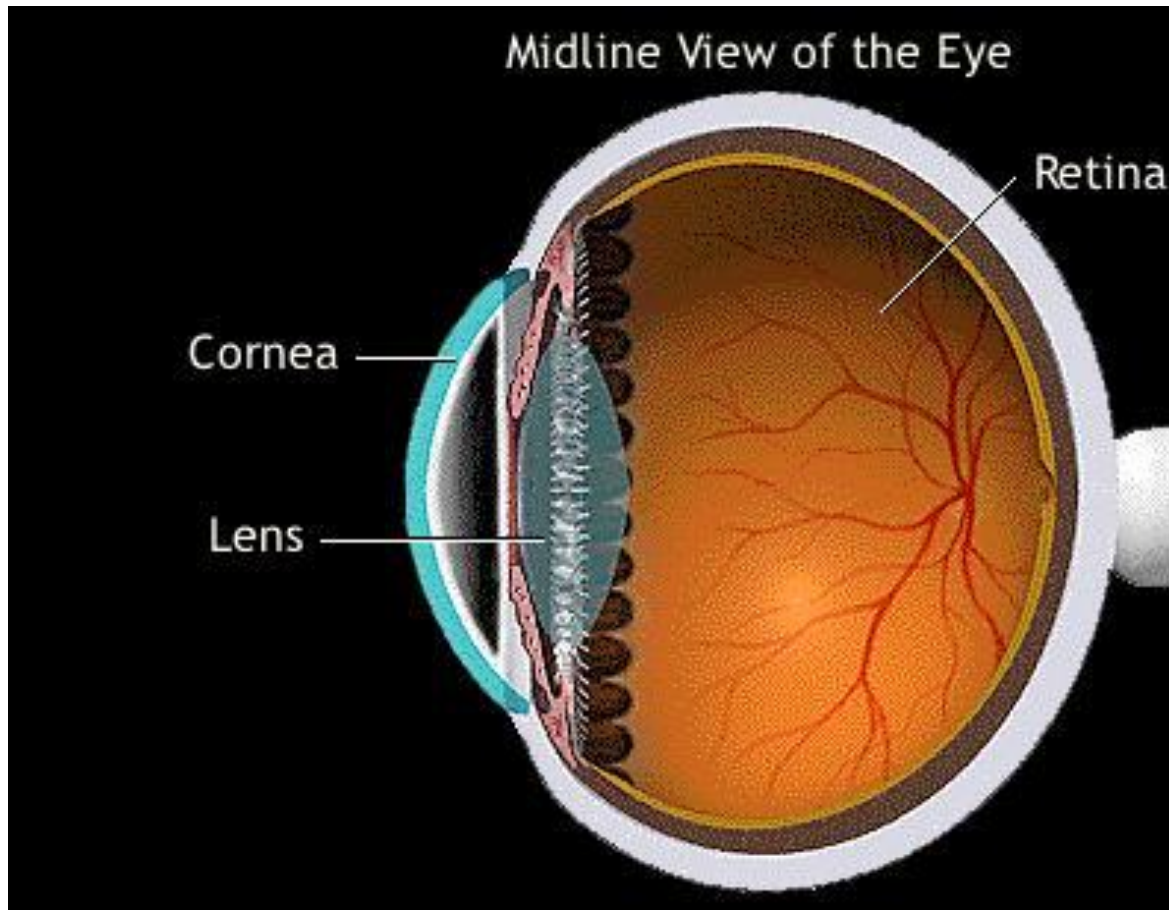
Human and Image Perception

Eye Basic Anatomy: Eye can be “extracted” and “disassembled” (for study)



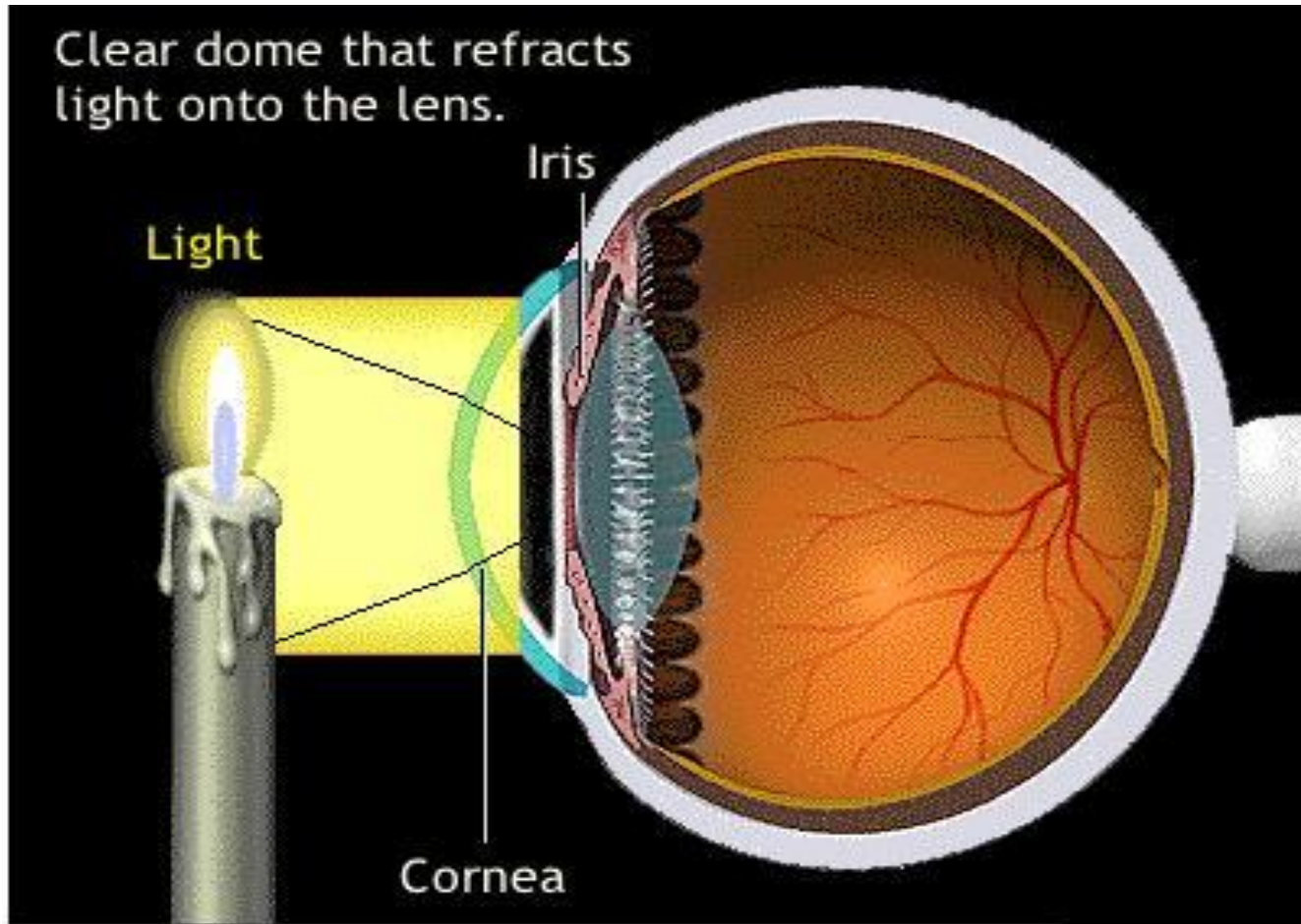
Human and Image Perception

Simplified Midline View of the Eye (To explain how eye focuses)

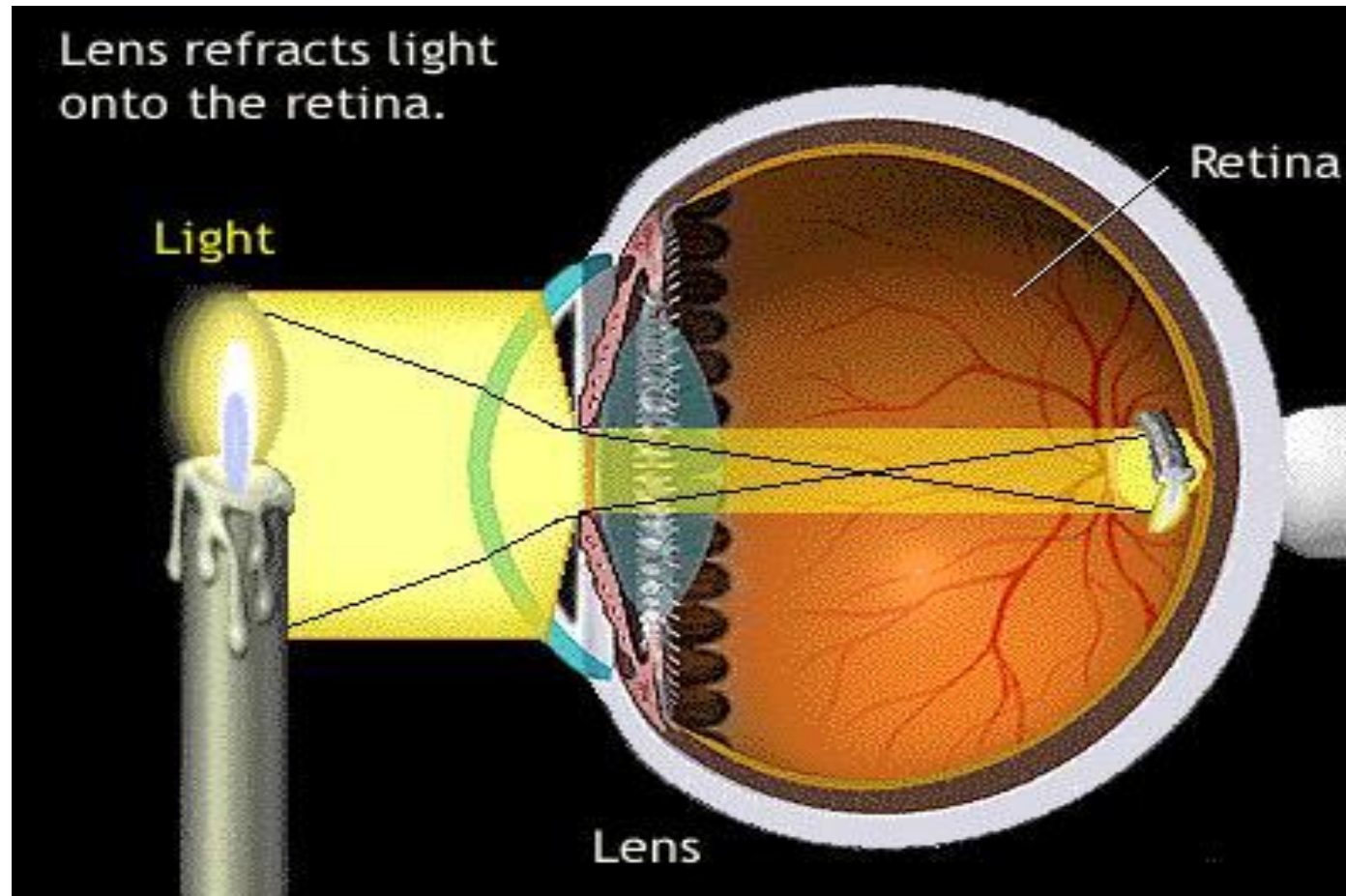


Human and Image Perception

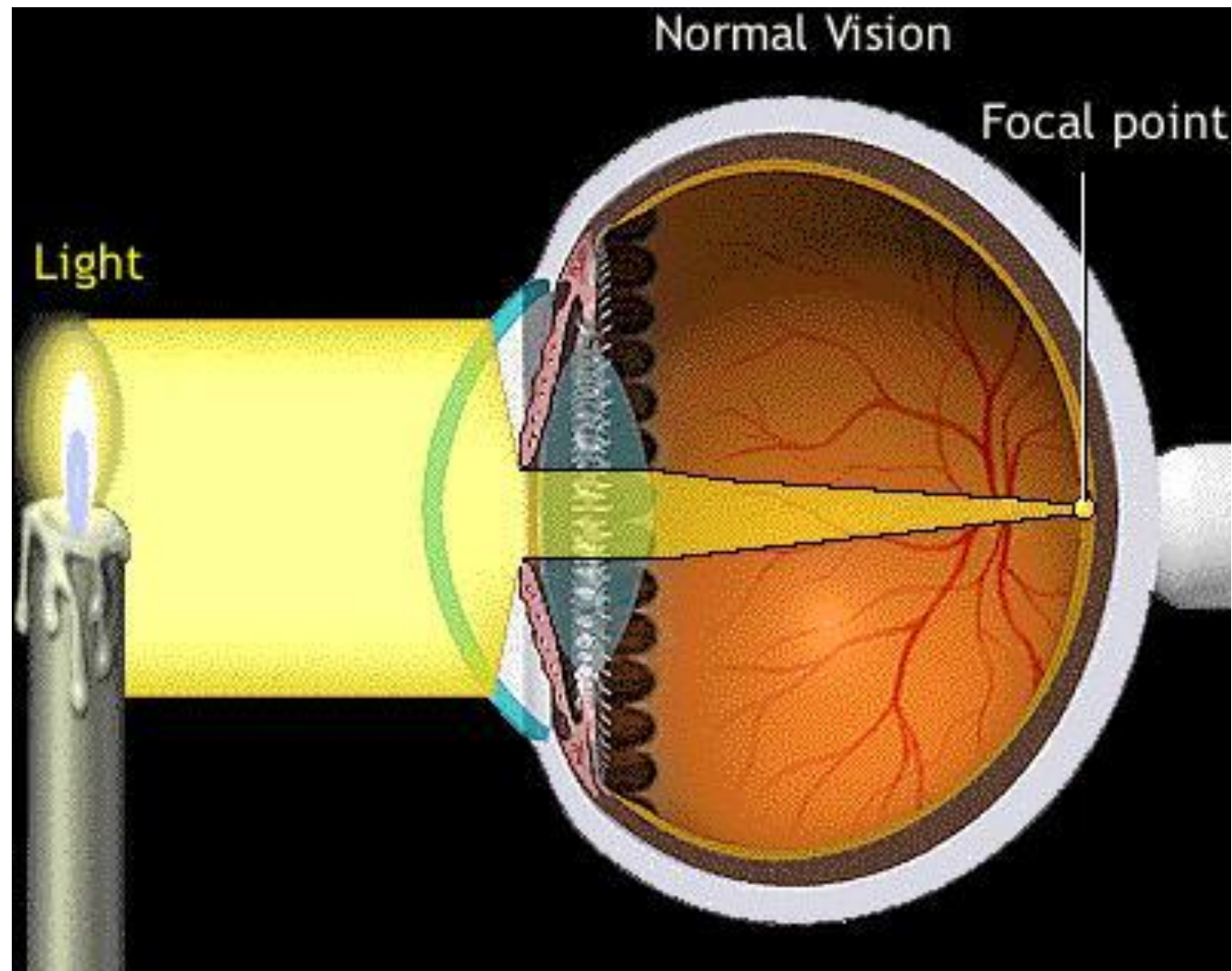
How eye focuses



Human and Image Perception



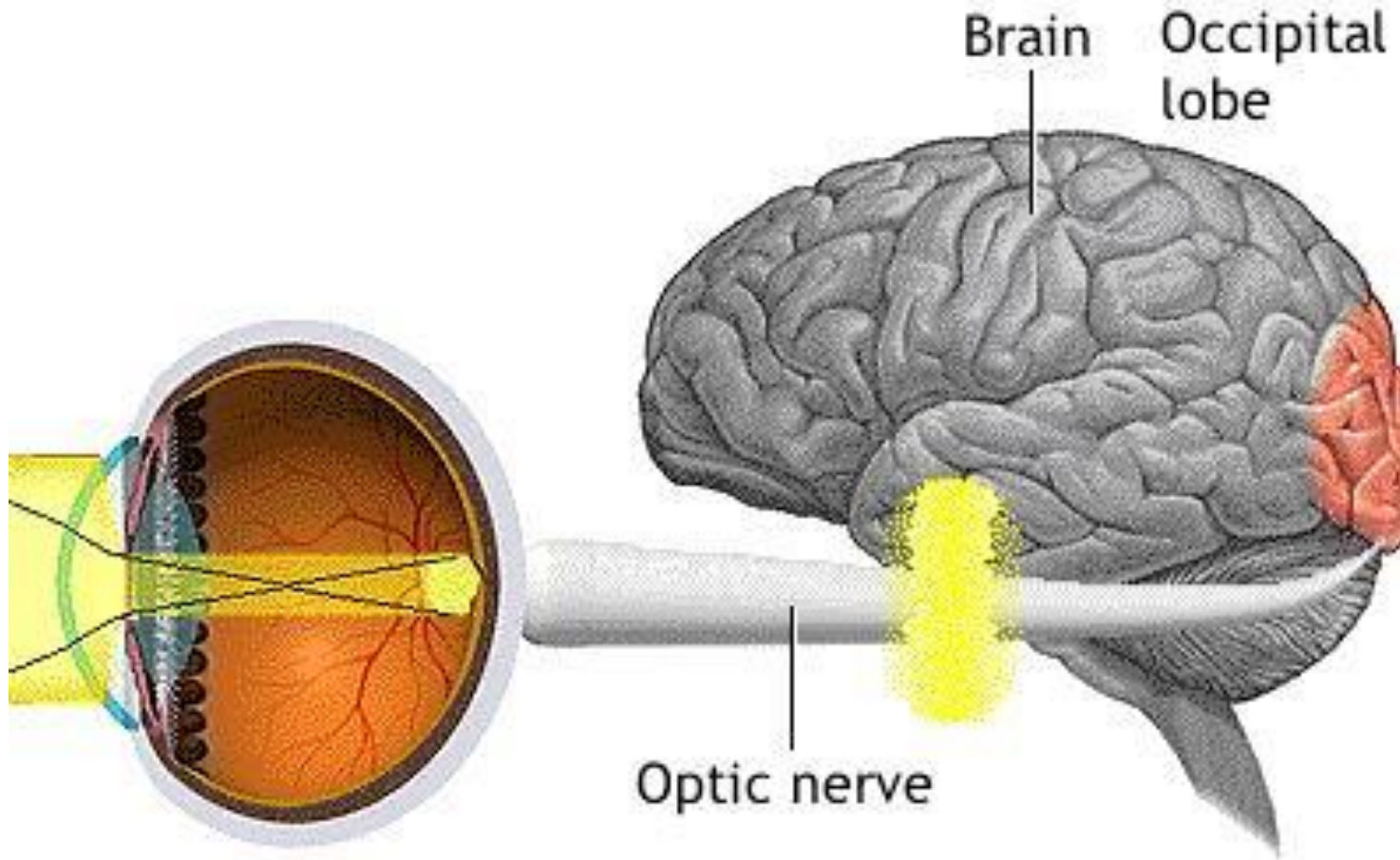
How eye focuses: Normal Vision-Focusing on the Retina



From Eye to the Brain

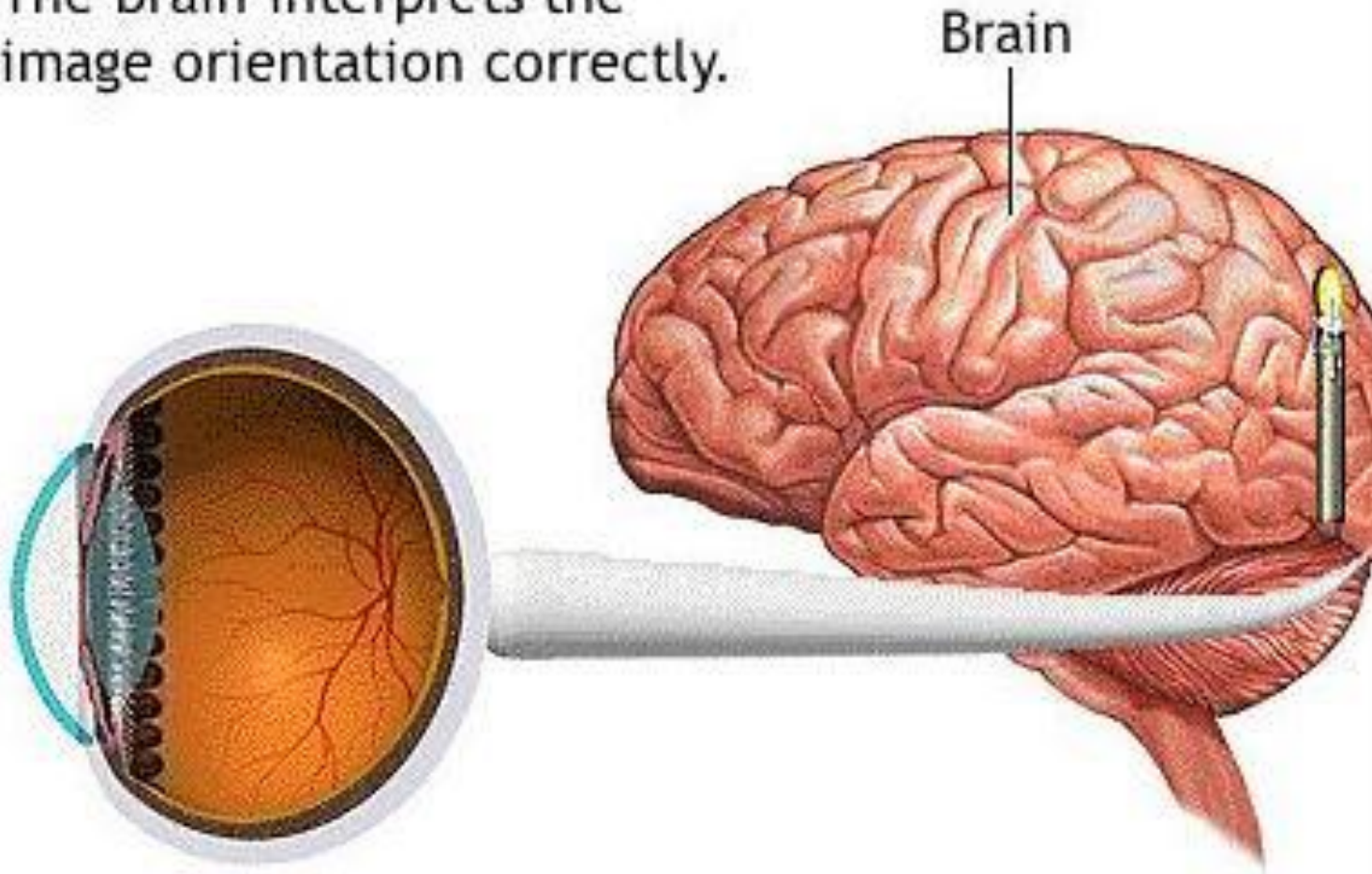
How does our brain receive this information?

Once the image is clearly focused on the sensitive part of the retina, energy in the light that makes up that image creates an electrical signal. Nerve impulses can then carry information about that image to the brain through the optic nerve.



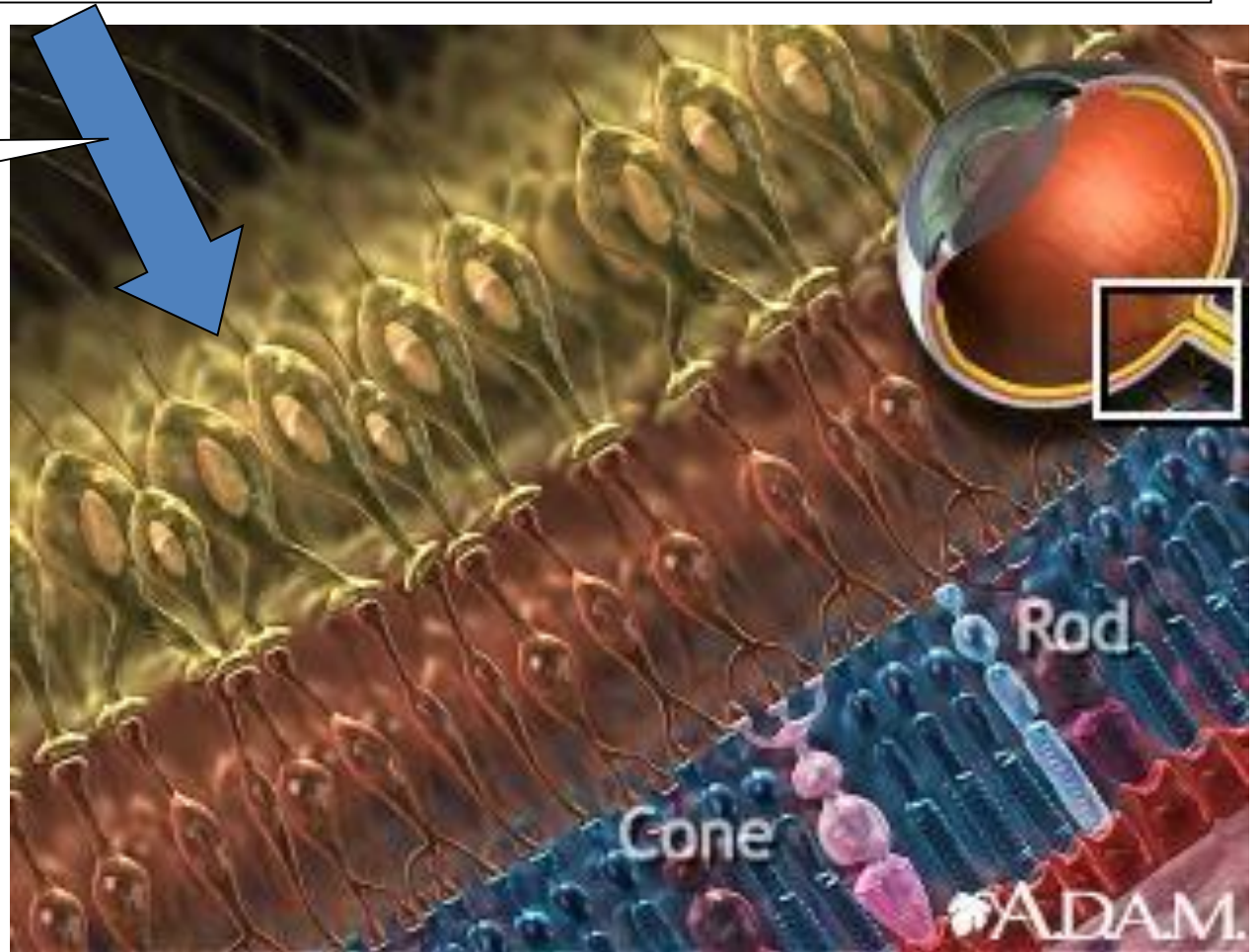
What we really “see” (after a number of days after the birth)

The brain interprets the image orientation correctly.



Retina: Contains specialized cells: (photo) receptors {*converts light into electrical signals*}
Rods – Black & White (Gray) images in low light (night)
Cones – Color Vision in bright light (day)

Direction of
Light

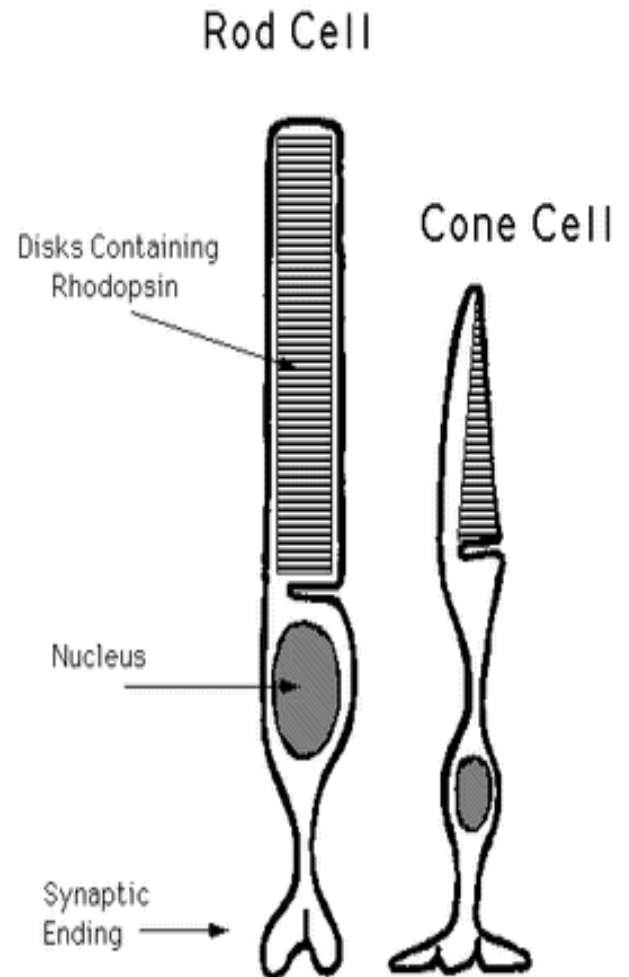


What happens when light reaches the retina?

It is packed with photosensitive cells called rods and cones.

- Cones are the cells responsible for daylight vision there are three different kinds – each responding to a different wavelength of light
- One responds to red light, one to green light and one to blue light.
- It is these cones which allow us to see in colour and detail
- Rods, on the other hand, are responsible for night vision.
- They are sensitive to light but not to wavelength information (colour)

In darkness, the cones do not function at all—so we need rods in order to see things even if it is only in shades of grey



Rods and Cones

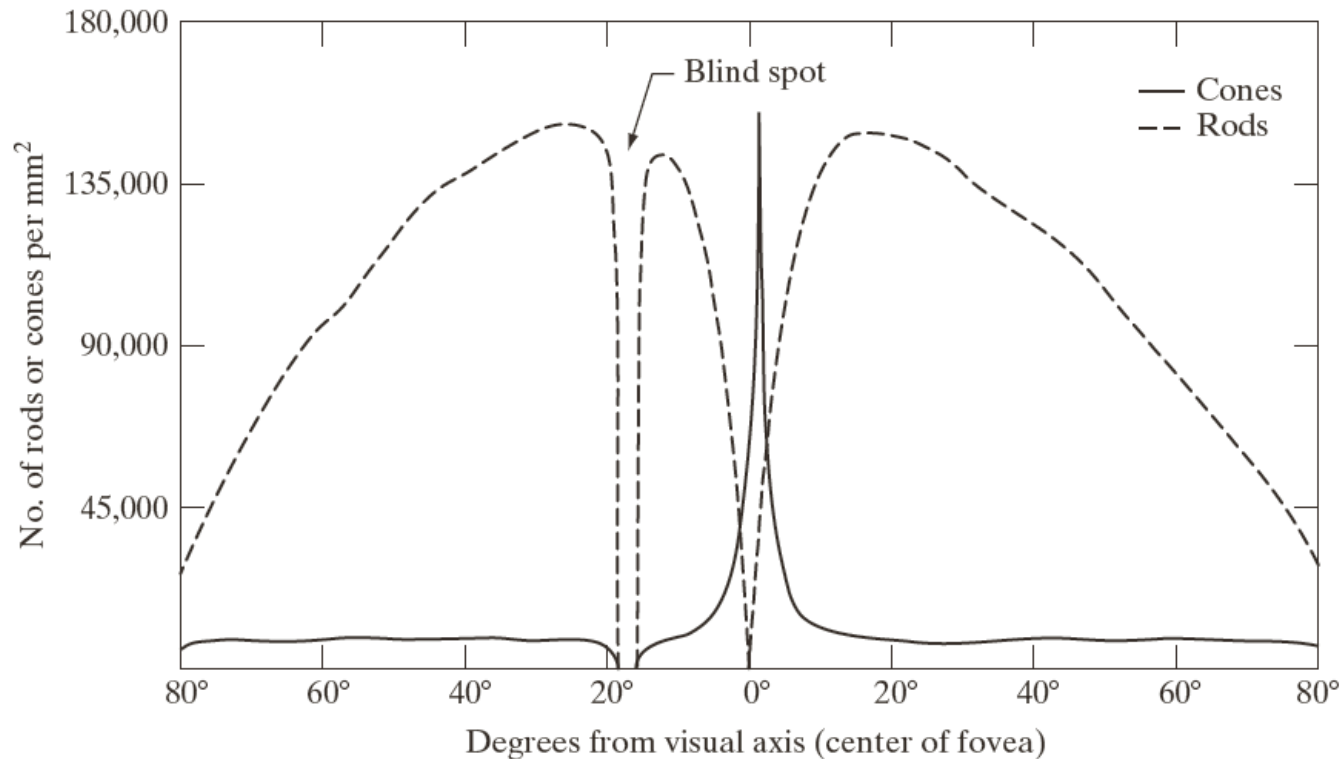
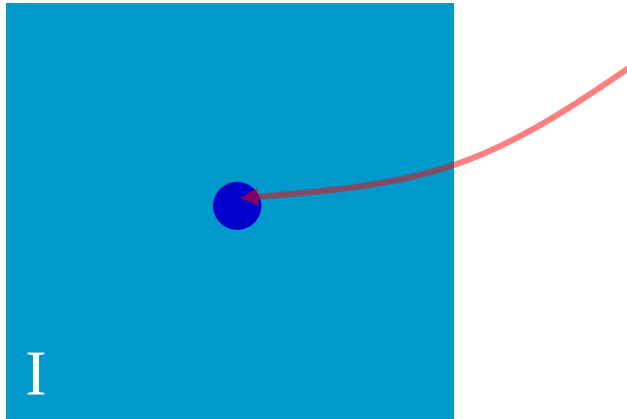


FIGURE 2.2
Distribution of
rods and cones in
the retina.

CONTRAST SENSITIVITY



Weber's ratio: $\Delta I_c/I$

Good brightness discrimination

$\Rightarrow \Delta I_c/I$ is small.

Bad brightness discrimination

$\Rightarrow \Delta I_c/I$ is large.

- The ability of the eye to discrimination b/w changes in brightness at any specific adaptation level is of considerable interest.
- I is uniformly illuminated on a flat area large enough to occupy the entire field of view.
- ΔI_c is the change in the object brightness required to just distinguish the object from the background

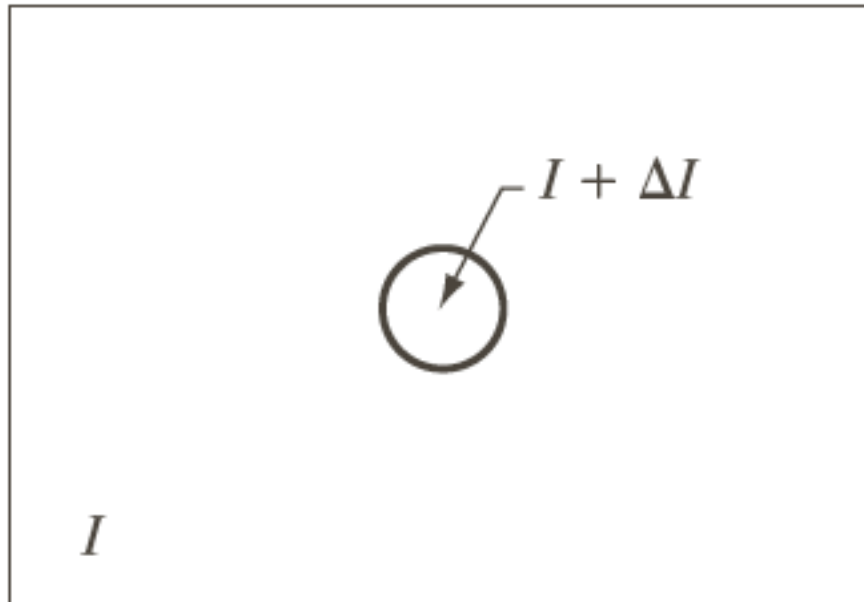


FIGURE 2.5 Basic experimental setup used to characterize brightness discrimination.

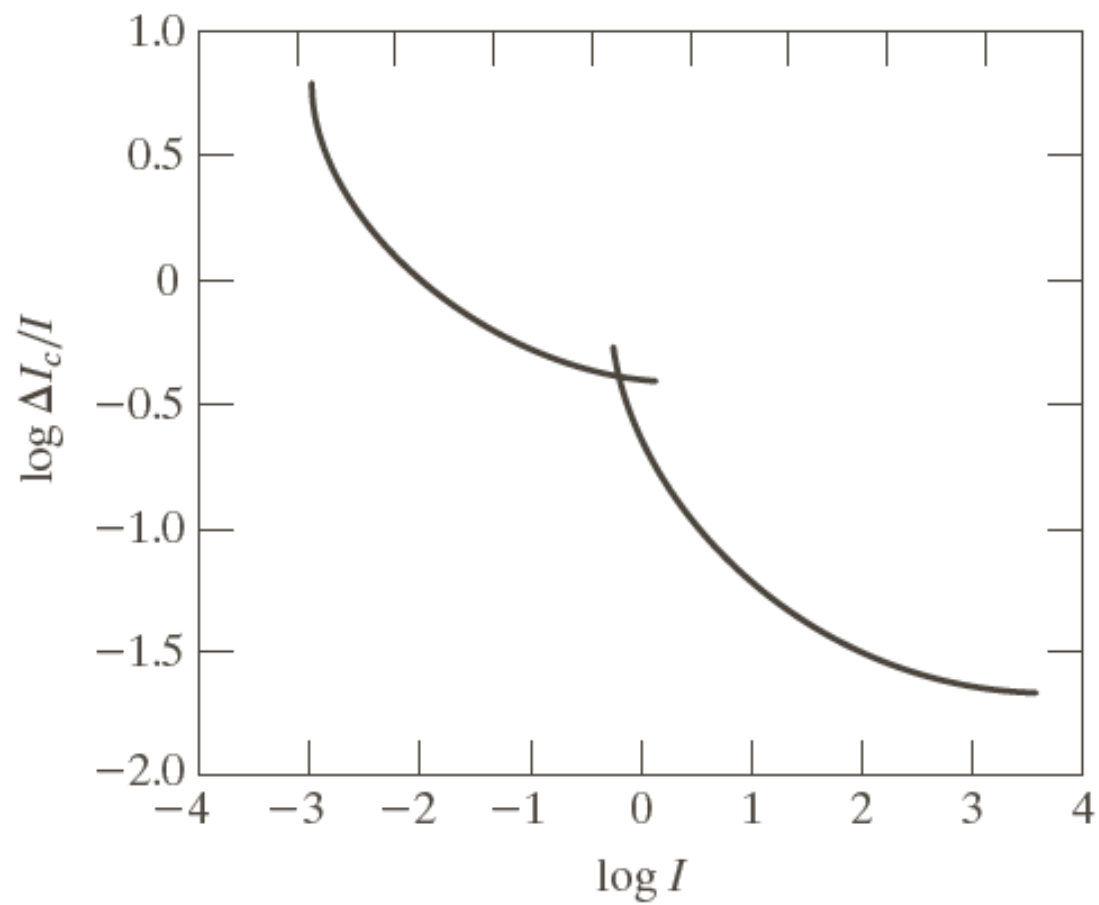


FIGURE 2.6
Typical Weber
ratio as a function
of intensity.

IMAGE FORMATION MODEL

- Image refers to a 2d light-intensity function, $f(x, y)$
- The amplitude of f at spatial coordinates (x, y) gives the intensity (brightness) of the image at that point.
- Light is a form of energy thus $f(x, y)$ must be nonzero and finite.

$$0 < f(x, y) < \infty.$$

IMAGE FORMATION MODEL

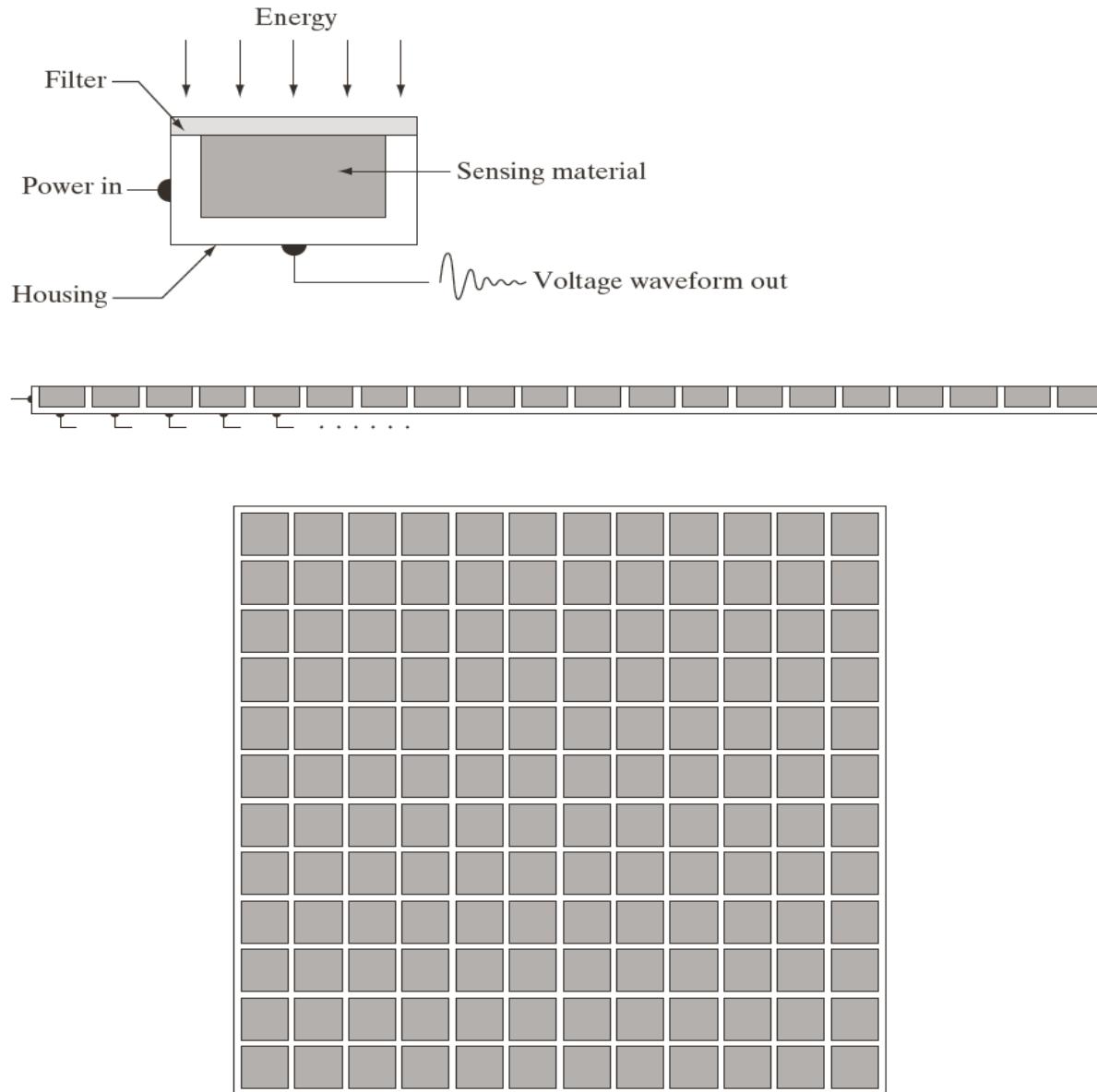
- The function $f(x, y)$ may be characterized by two components:
 - The amount of source light incident on the scene being viewed $\Rightarrow$ illumination.
 - The amount of light reflected by the objects in the scene $\Rightarrow$ reflectance.

$$f(x, y) = i(x, y)r(x, y)$$

$$0 < i(x, y) < \infty$$

$$0 < r(x, y) < 1.$$

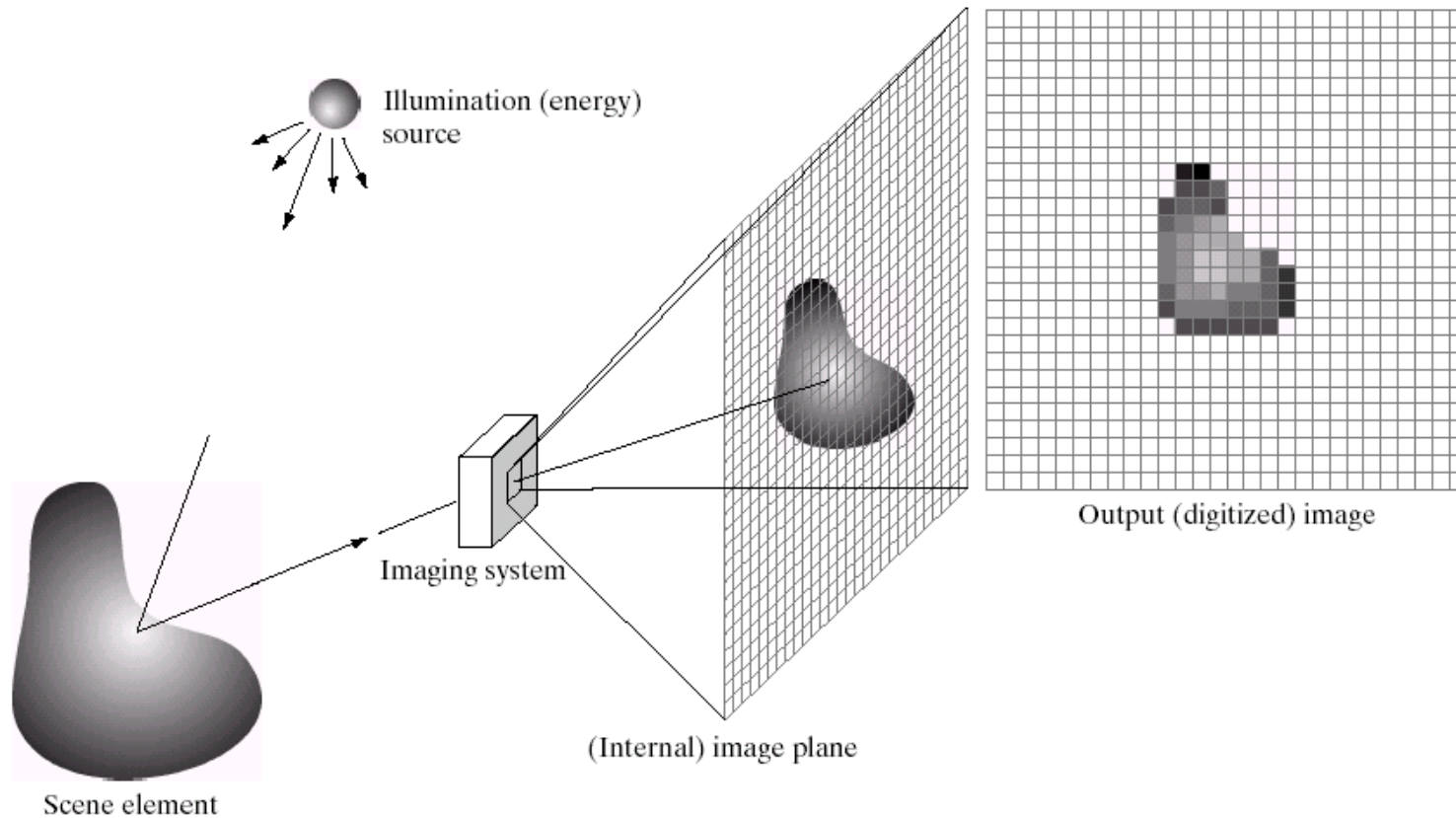
Image Acquisition



a
b
c

FIGURE 2.12
(a) Single imaging sensor.
(b) Line sensor.
(c) Array sensor.

Image Acquisition



a b c d e

FIGURE 2.15 An example of the digital image acquisition process. (a) Energy (“illumination”) source. (b) An element of a scene. (c) Imaging system. (d) Projection of the scene onto the image plane. (e) Digitized image.

Sampling and Quantization

- ◆ Sampling:

- Digitization of the spatial coordinates (x,y)

- ◆ Quantization:

- Digitization in amplitude (also known as gray level quantization)

Sampling and Quantization

◆ Quantization

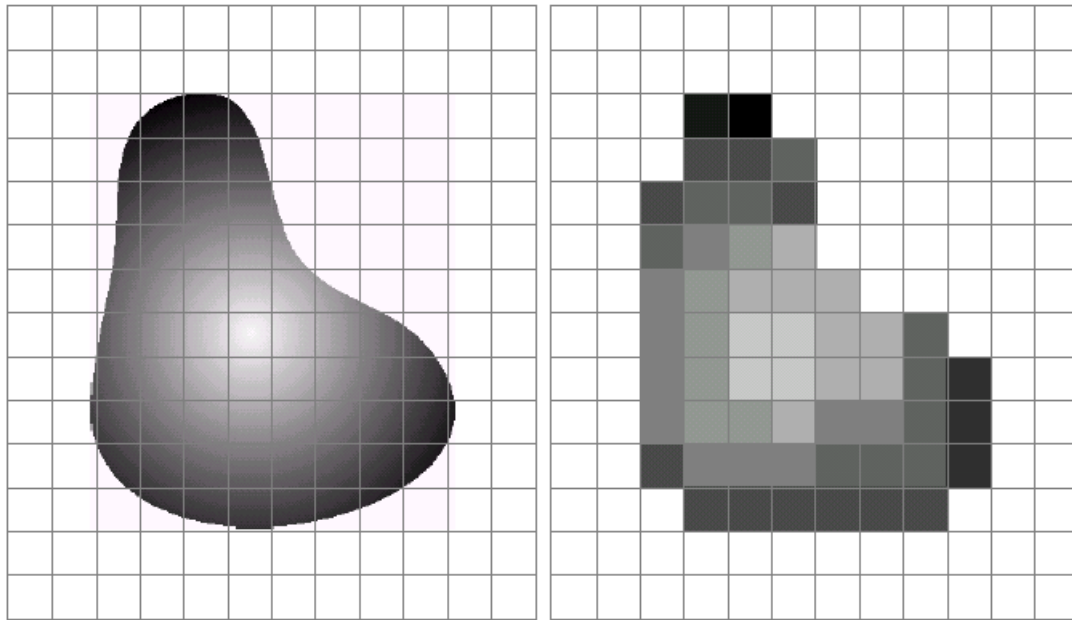
- 8 bit quantization: $2^8 = 256$ gray levels (0: black, 255: white)
- 1 bit quantization: 2 gray levels (0: black, 1: white) – binary

◆ Sampling

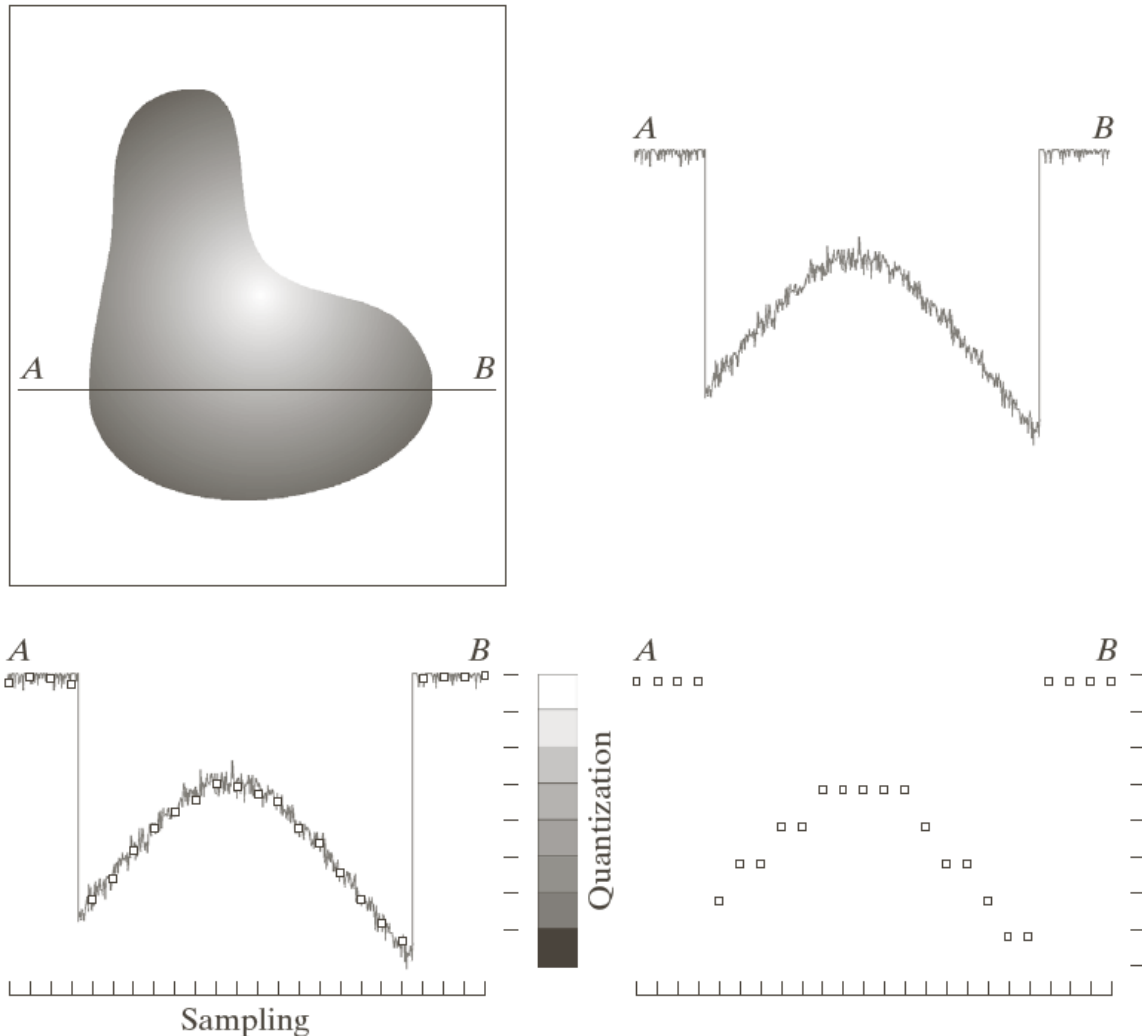
- Commonly used number of samples (resolution)
 - Digital still cameras: 640x480, 1024x1024, 4064 x 2704
 - Digital video cameras: 640x480 at 30 frames/second (fps)

Sampling and Quantization

- ◆ Digital Image is an approximation of a real world scene



Sampling and Quantization

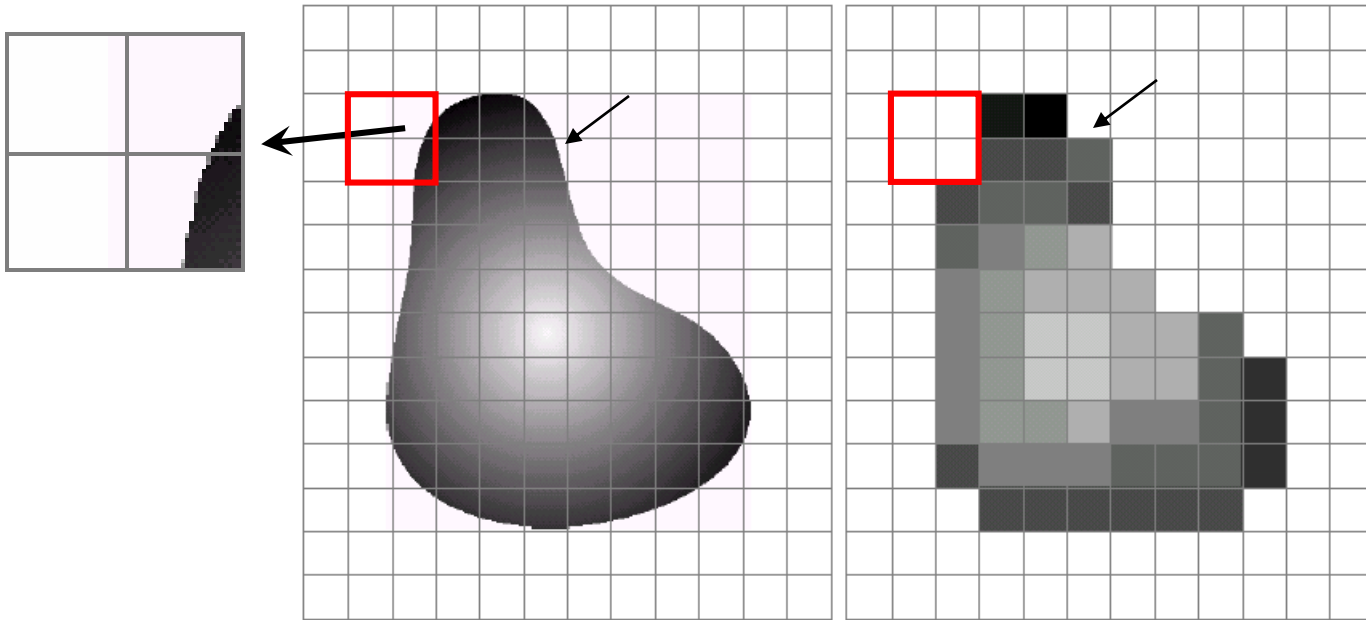


a	b
c	d

FIGURE 2.16
Generating a digital image.
(a) Continuous image. (b) A scan line from *A* to *B* in the continuous image, used to illustrate the concepts of sampling and quantization. (c) Sampling and quantization. (d) Digital scan line.

Image Formation

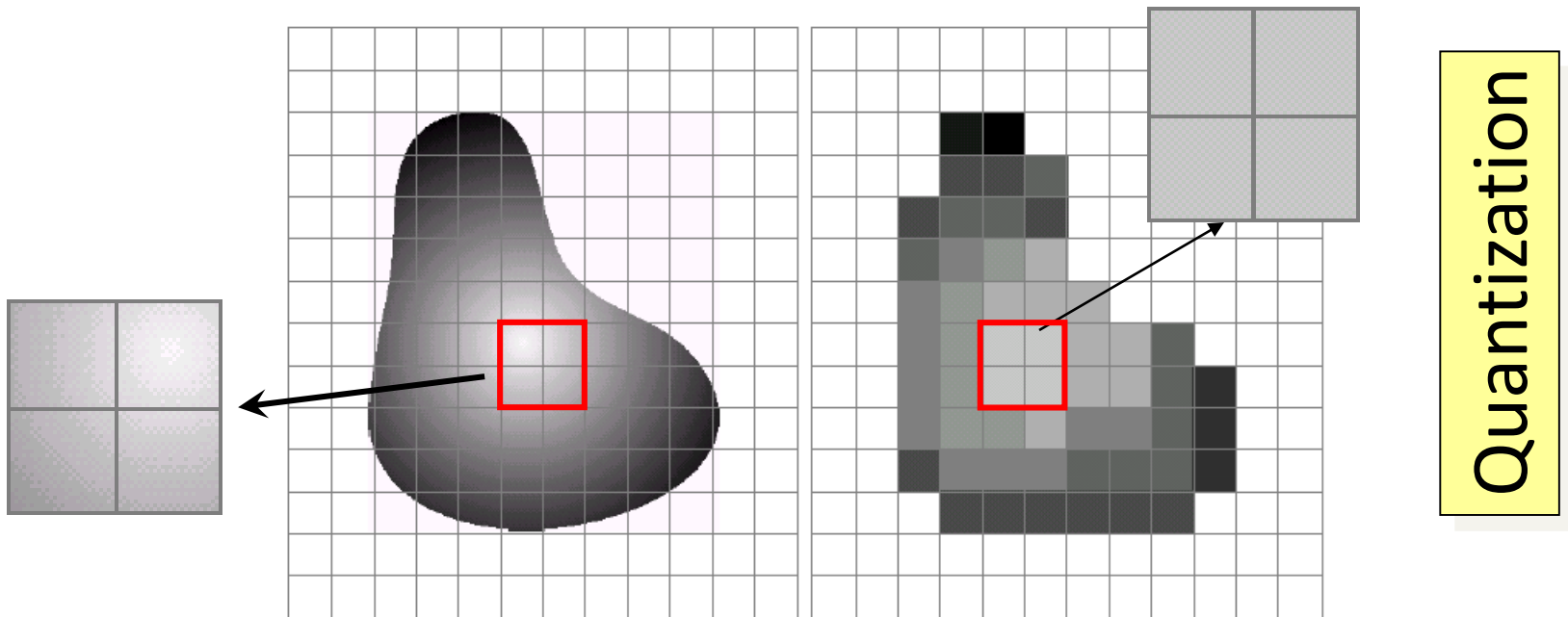
- ◆ Digital Image is an approximation of a real world scene



Sampling

Image Formation

- ◆ Digital Image is an approximation of a real world scene

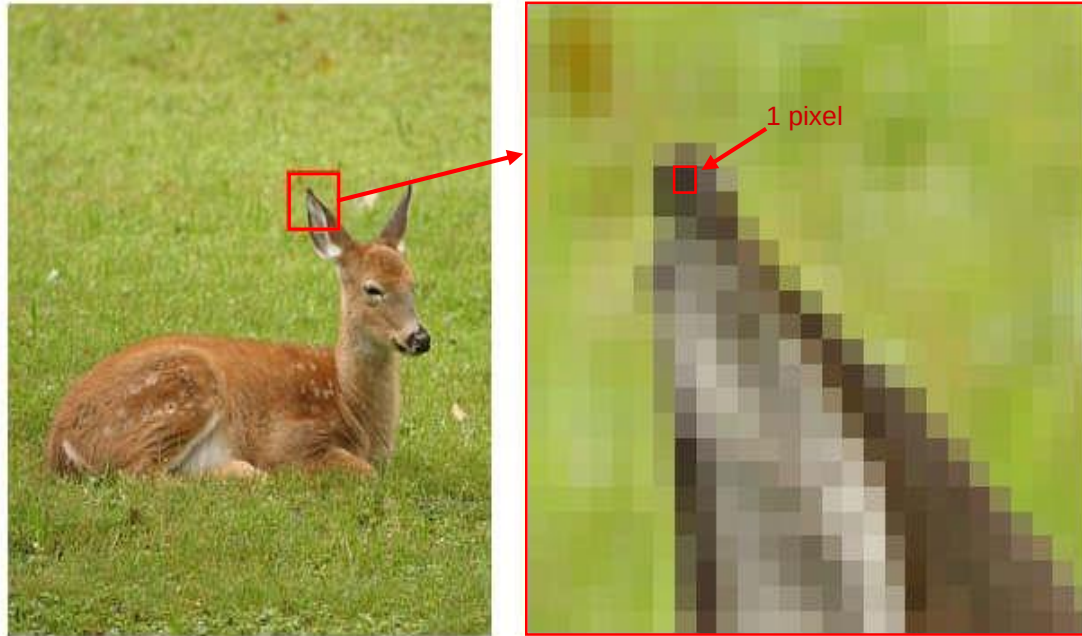


Digital Images

Digital Image

a grid of squares,
each of which
contains a single
color

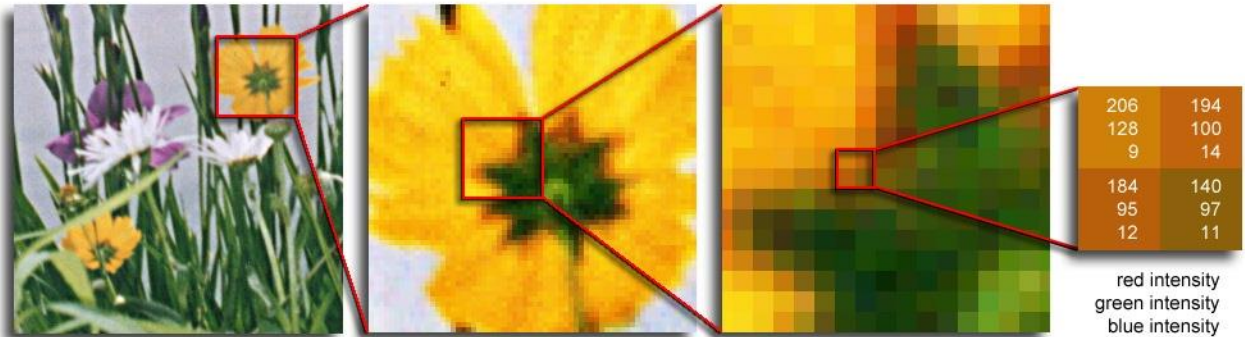
each square is
called a pixel (for
picture element)



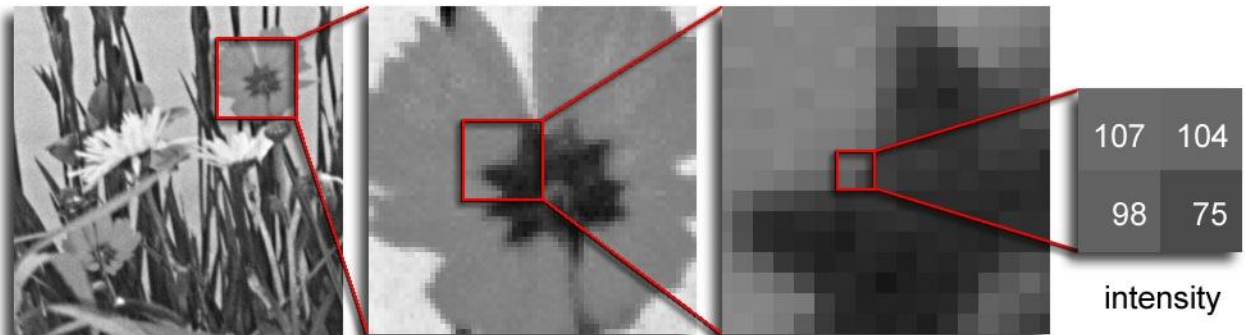
Digital Image

Color images have 3 values per pixel; monochrome images have 1 value per pixel.

a grid of squares, each of which contains a single color



each square is called a pixel (for *picture element*)

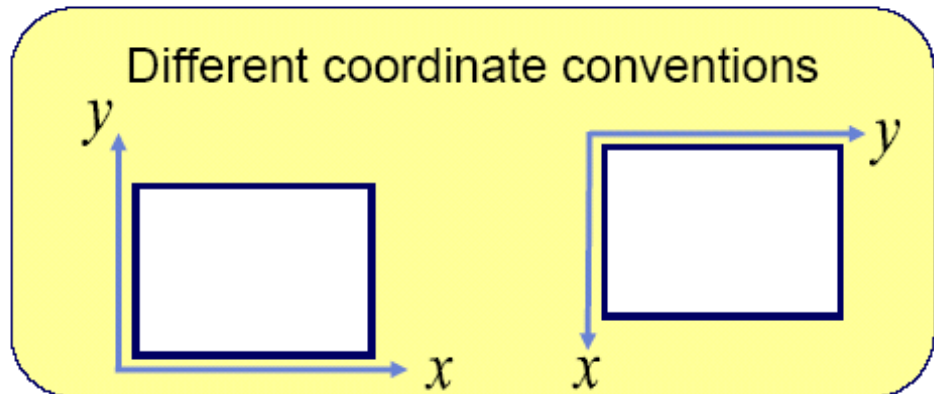
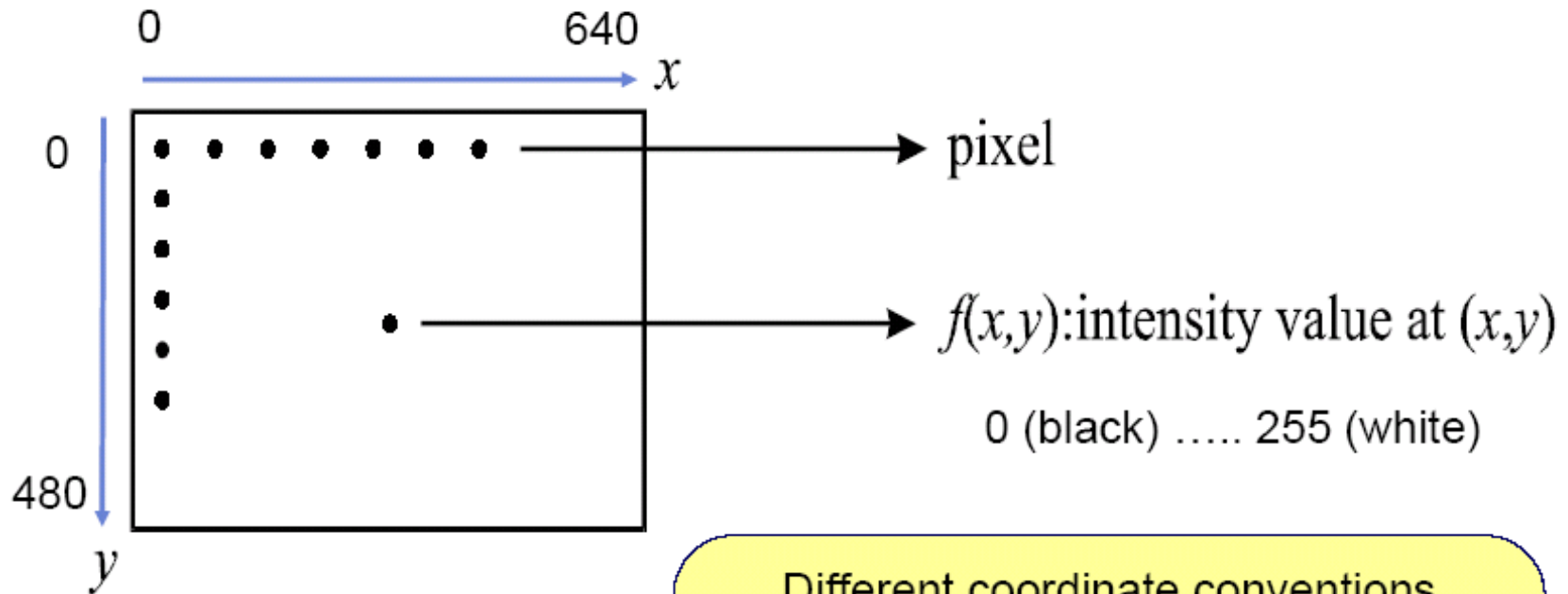


Digital Image

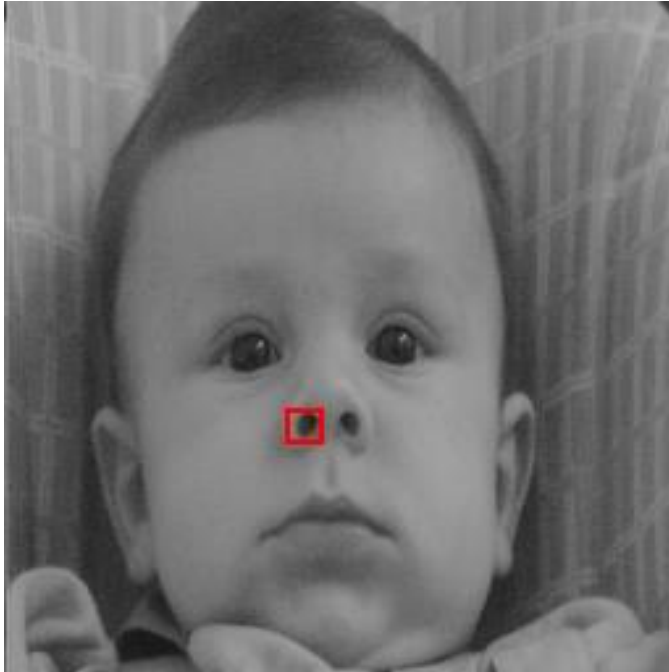
- A set of pixels (picture elements, pels)
- Pixel means
 - pixel coordinate
 - pixel value
 - or both
- Both coordinates and value are discrete

Example

640 x 480 8-bit image



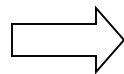
DIGITAL IMAGE REPRESENTATION



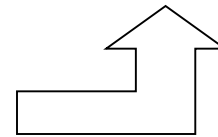
PIXEL VALUES IN HIGHLIGHTED REGION

99	71	61	51	49	40	35	53	86	99
93	74	53	56	48	46	48	72	85	102
101	69	57	53	54	52	64	82	88	101
107	82	64	63	59	60	81	90	93	100
114	93	76	69	72	85	94	99	95	99
117	108	94	92	97	101	100	108	105	99
116	114	109	106	105	108	108	102	107	110
115	113	109	114	111	111	113	108	111	115
110	113	111	109	106	108	110	115	120	122
103	107	106	108	109	114	120	124	124	132

CAMERA



DIGITIZER



**A set of number
in 2D grid**

Samples the analog data and digitizes it.

What is a Digital Image? (cont...)

- Common image formats include:
 - 1 sample per point (B&W or Grayscale)
 - 3 samples per point (Red, Green, and Blue)



- For most of this course we will focus on grey-scale images

GRAY LEVEL

- WE CALL THE INTENSITY OF A MONOCHROME IMAGE f AT COORDINATE (x, y) THE GRAY LEVEL (L) OF THE IMAGE AT THAT POINT.
- Thus, L lies in the range

$$L_{\min} \leq \ell \leq L_{\max}$$

- $L_{\min}$ is positive and $L_{\max}$ is finite.
- Gray scale = $[L_{\min}, L_{\max}]$
- Common practice, shift the interval to $[0, L]$: 0 = black, $L-1$ = white

Digital Image Representation

- ◆ Image Size

- Number of bits required to store an image

$$b = M \times N \times k$$

- Image having 2^k intensity levels
 - k – bit image
 - 256 intensity levels – 8 bit image

Image Size

TABLE 2.1

Number of storage bits for various values of N and k .

N/k	1 ($L = 2$)	2 ($L = 4$)	3 ($L = 8$)	4 ($L = 16$)	5 ($L = 32$)	6 ($L = 64$)	7 ($L = 128$)	8 ($L = 256$)
32	1,024	2,048	3,072	4,096	5,120	6,144	7,168	8,192
64	4,096	8,192	12,288	16,384	20,480	24,576	28,672	32,768
128	16,384	32,768	49,152	65,536	81,920	98,304	114,688	131,072
256	65,536	131,072	196,608	262,144	327,680	393,216	458,752	524,288
512	262,144	524,288	786,432	1,048,576	1,310,720	1,572,864	1,835,008	2,097,152
1024	1,048,576	2,097,152	3,145,728	4,194,304	5,242,880	6,291,456	7,340,032	8,388,608
2048	4,194,304	8,388,608	12,582,912	16,777,216	20,971,520	25,165,824	29,369,128	33,554,432
4096	16,777,216	33,554,432	50,331,648	67,108,864	83,886,080	100,663,296	117,440,512	134,217,728
8192	67,108,864	134,217,728	201,326,592	268,435,456	335,544,320	402,653,184	469,762,048	536,870,912

Spatial & Gray Level Resolution

Spatial Resolution



1024



512



256



128



32

64

Spatial Resolution



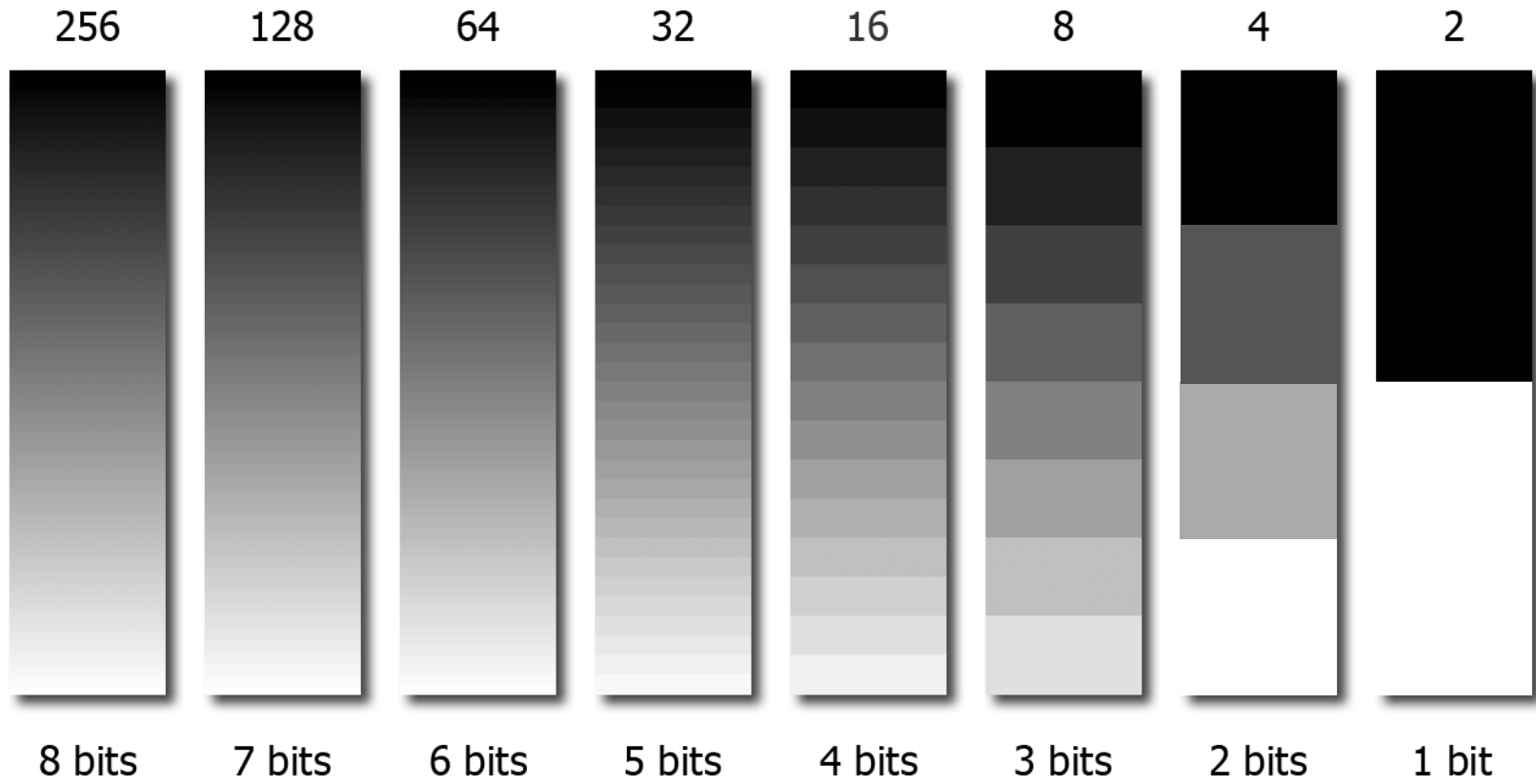
Intensity Level Resolution

- ◆ *Intensity level resolution* refers to the number of intensity levels used to represent the image
 - The more intensity levels used, the finer the level of detail in an image
 - Intensity level resolution is usually given in terms of the number of bits used to store each intensity level

Intensity Level Resolution

Number of Bits	Number of Intensity Levels	Examples
1	2	0, 1
2	4	00, 01, 10, 11
4	16	0000, 0101, 1111
8	256	00110011, 01010101
16	65,536	1010101010101010

Intensity Level Resolution



Intensity Level Resolution

256 grey levels (8 bits per pixel)



128 grey levels (7 bpp)



64 grey levels (6 bpp)



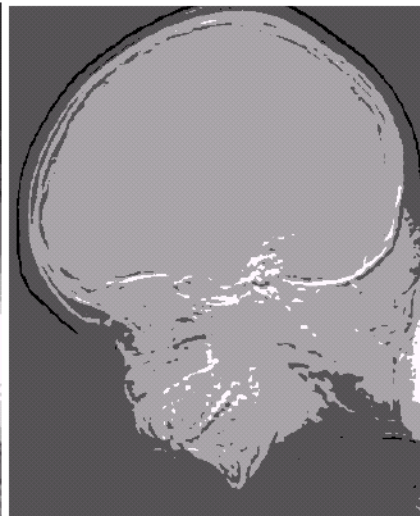
32 grey levels (5 bpp)



16 grey levels (4 bpp)



8 grey levels (3 bpp)



4 grey levels (2 bpp)



2 grey levels (1 bpp)

Resolution: How much is enough?

- ◆ How many samples and gray levels are required for a good approximation?
 - Quality of an image depends on number of pixels and gray-level number
 - The more these parameters are increased, the closer the digitized array approximates the original image
 - But: Storage & processing requirements increase rapidly as a function of N , M , and k

Resolution: How much is enough?

- ◆ Depends on what is in the image and what you would like to do with it



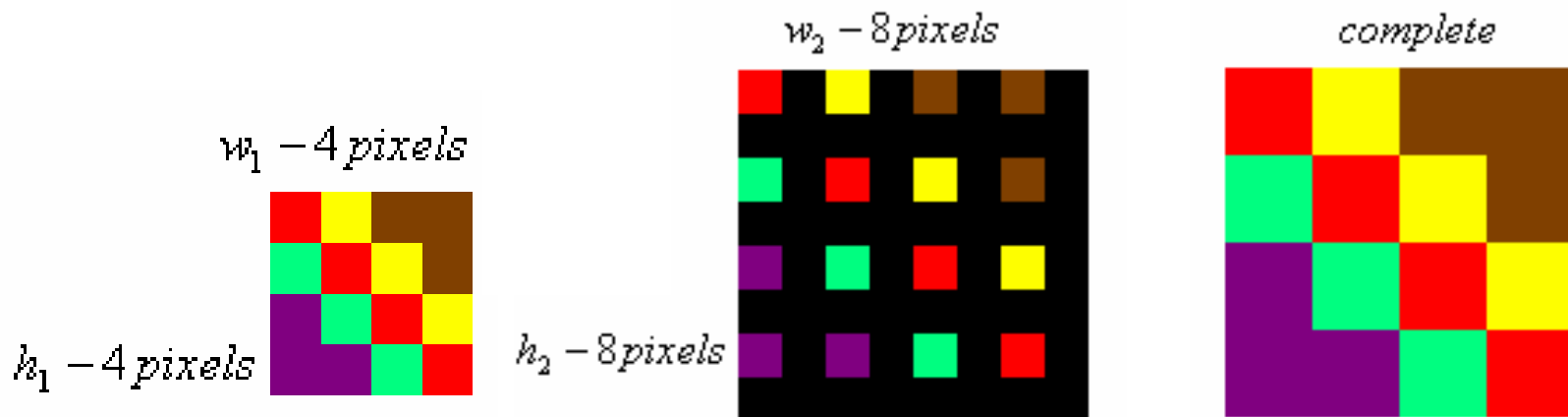
Image Resizing

- ◆ Pixel replication

[1 2 3 4 5]

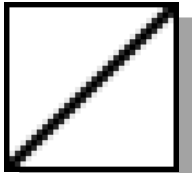
[1 1 2 2 3 3 4 4 5 5] (One step)

[1 1 1 2 2 2 3 3 3 4 4 4 5 5 5] (Two step)

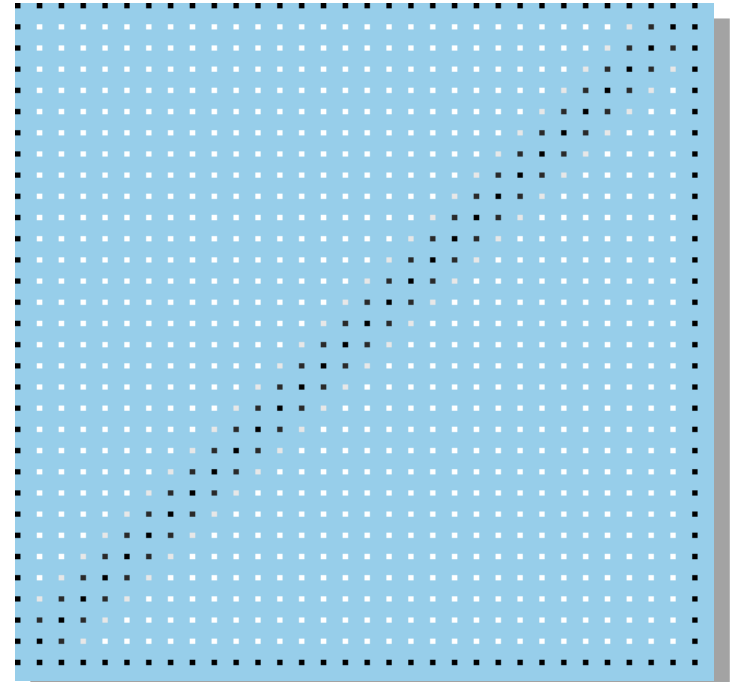


Enlarging an Image

Example:
zoom this
image 4x to
get this
image.

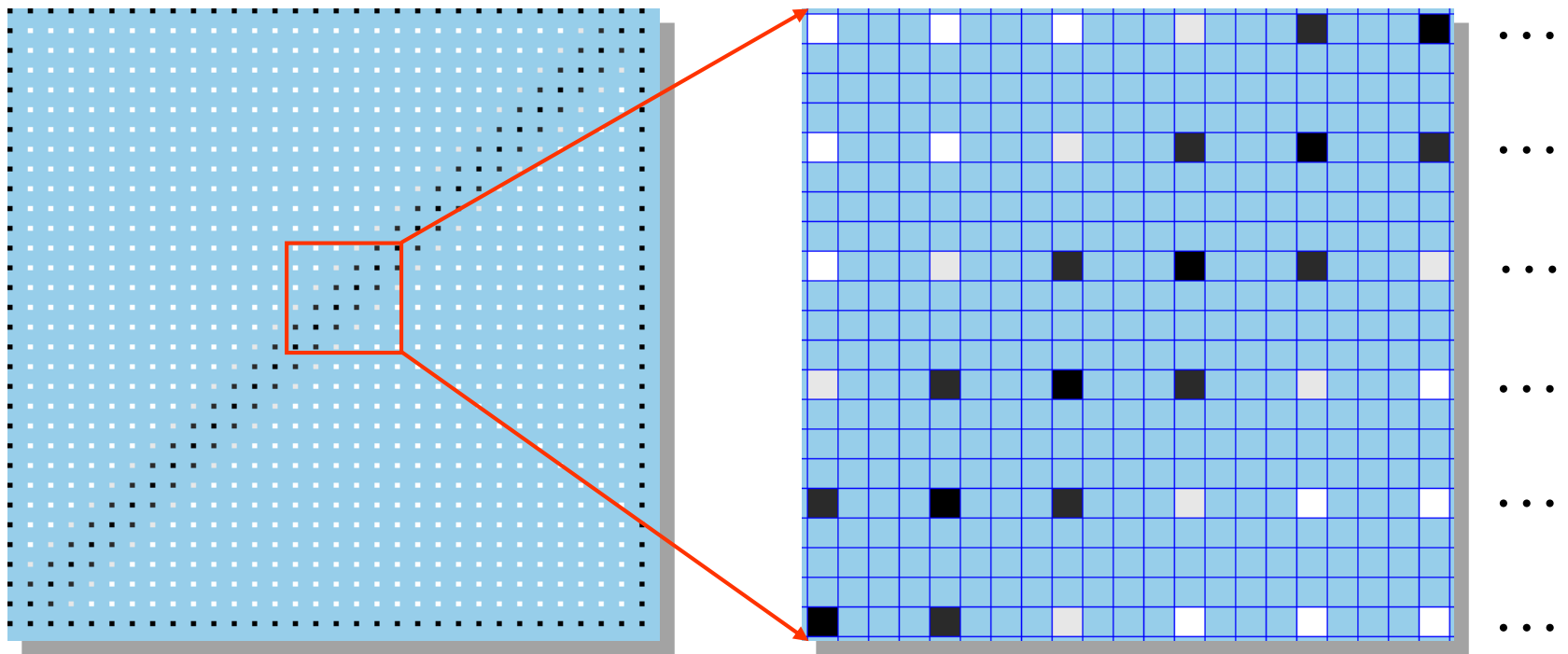


Start with a blank image 4 times the
linear dimensions of the original.



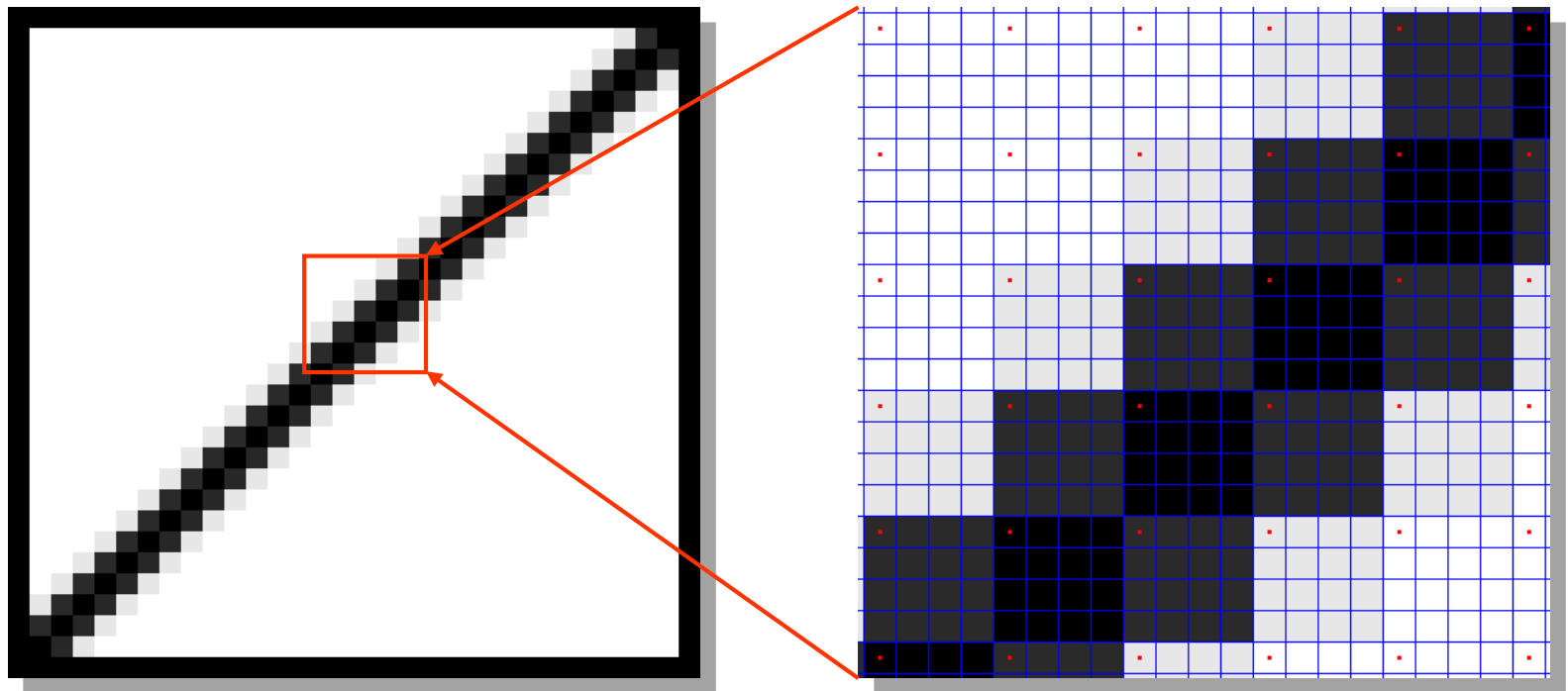
Fill in every 4th pixel in every 4th
row with the original pixel values.

Enlarging an Image



Detail showing every 4th pixel in every 4th row with the original pixel values.

Enlarging an Image



Replicate the values

Image Interpolation

- ◆ Nearest neighbour interpolation
 - Simple but produces undesired artefacts
- ◆ Bilinear Interpolation
 - Contribution from 4 neighbors
- ◆ Bicubic Interpolation
 - Contribution from 16 neighbors



Interpolation: Comparison

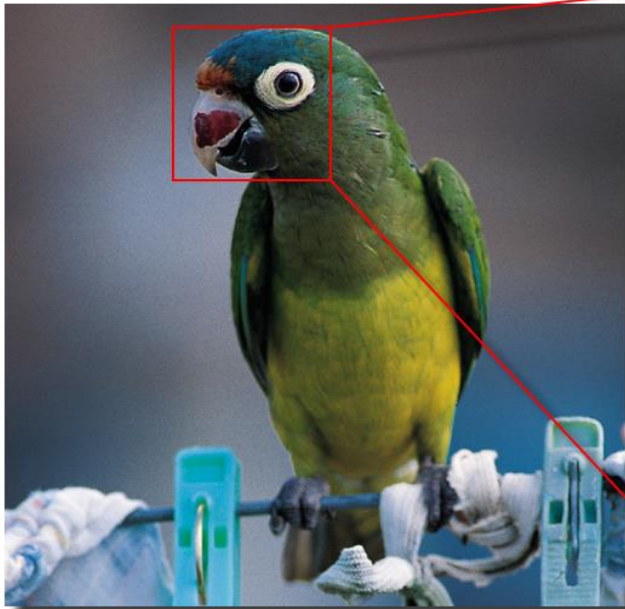


We'll enlarge this image by a factor of 4 ...

... via bilinear interpolation and compare it to a nearest neighbor enlargement.

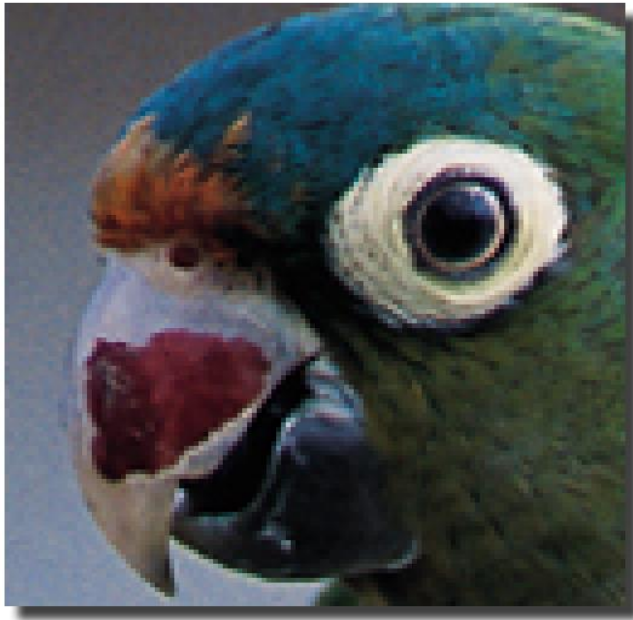
Interpolation: Comparison

Original
Image



To better see what happens, we'll look at the parrot's eye.

Interpolation: Comparison

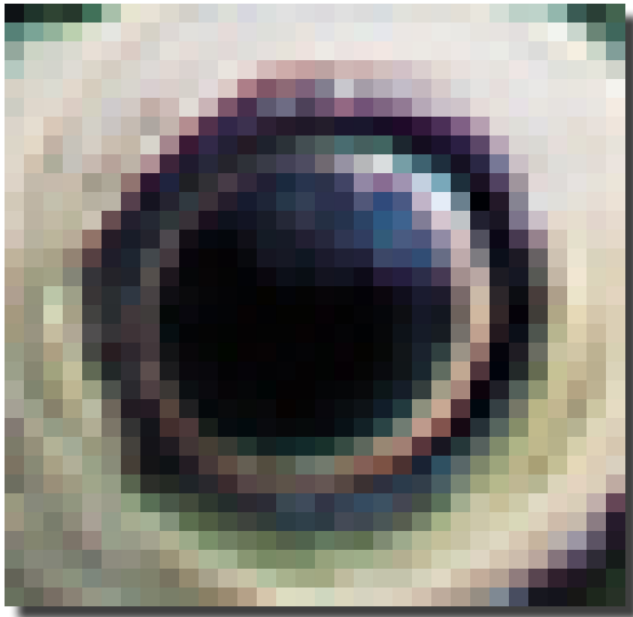


Pixel replication



Bilinear interpolation

Interpolation: Comparison



Pixel replication



Bilinear interpolation

Readings from Book (3rd Edn.)

- Chapter – 2

Read topics from 2.1 to 2.4 from book



Acknowledgements

- ◆ Statistical Pattern Recognition: A Review – A.K Jain et al., PAMI (22) 2000
- ◆ Pattern Recognition and Analysis Course – A.K. Jain, MSU
- ◆ *Pattern Classification*” by Duda et al., John Wiley & Sons.
- ◆ Digital Image Processing”, Rafael C. Gonzalez & Richard E. Woods, Addison-Wesley, 2002
- ◆ Machine Vision: Automated Visual Inspection and Robot Vision”, David Vernon, Prentice Hall, 1991
- ◆ www.eu.aibo.com/
- ◆ Advances in Human Computer Interaction, Shane Pinder, InTech, Austria, October 2008